



InsightIQ

Ask your data why something changed

From alert to answer.

Automated root-cause analysis, **native to ClickHouse**.

Click-a-thon 2026 · InMobi Track

Team 404Duos

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`insight-iq-woad.vercel.app`

THE PROBLEM

Alerts tell you what moved. Nobody tells you why.

01

Hours, not seconds

Analysts manually slice by geo, device, format, and advertiser to chase a single revenue dip.

02

Fragmented tooling

Dashboards show the drop; root-cause still lives in someone's head or a Slack thread.

03

Scale breaks the habit

At InMobi scale, thousands of advertiser × metric combinations move every day — manual triage doesn't scale.

THE SOLUTION

Detect → drill down → diagnose.
In seconds, not hours.

● Detect

ClickHouse-native cascade — seasonality baseline, noise-floored Z-score.

● Drill down

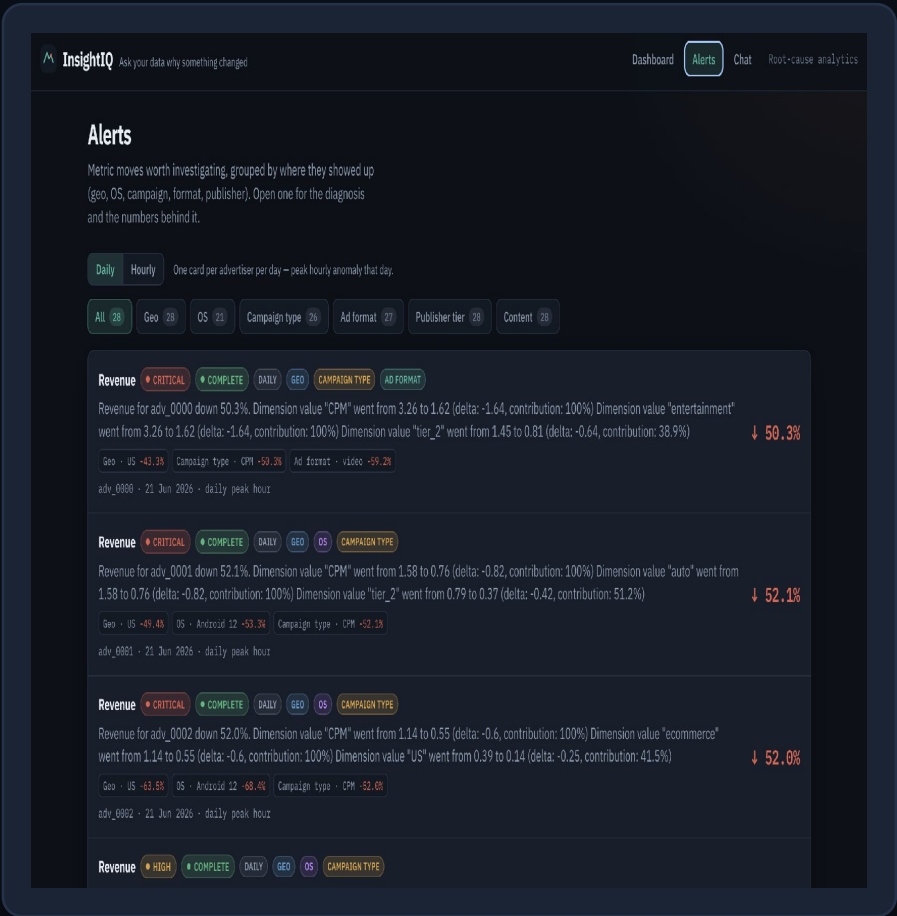
Multi-dimension contribution analysis pinpoints the responsible segment.

● Diagnose

Deterministic diagnosis with citations — Gemini narrates evidence only.

● Trust

Langfuse traces every chat / investigate / narrate turn end-to-end.



Live alert wall — critical revenue drops, ranked

HOW IT WORKS

Five deterministic steps, one honest answer

1

Seasonality baseline

Same hour × day-of-week, averaged over the prior ~4 weeks — a weekend never looks like an incident.

2

Noise-floored Z-score

stddev floored at 0.05 so penny-level noise can't explode into a false alarm; alert when $|z| > 3$.

3

Contribution analysis

Ranks which geo / OS / format / publisher / campaign segment is actually driving the move.

4

Ruled-out honesty

Seasonality and non-causal factors are explicitly marked ruled-out, not hidden.

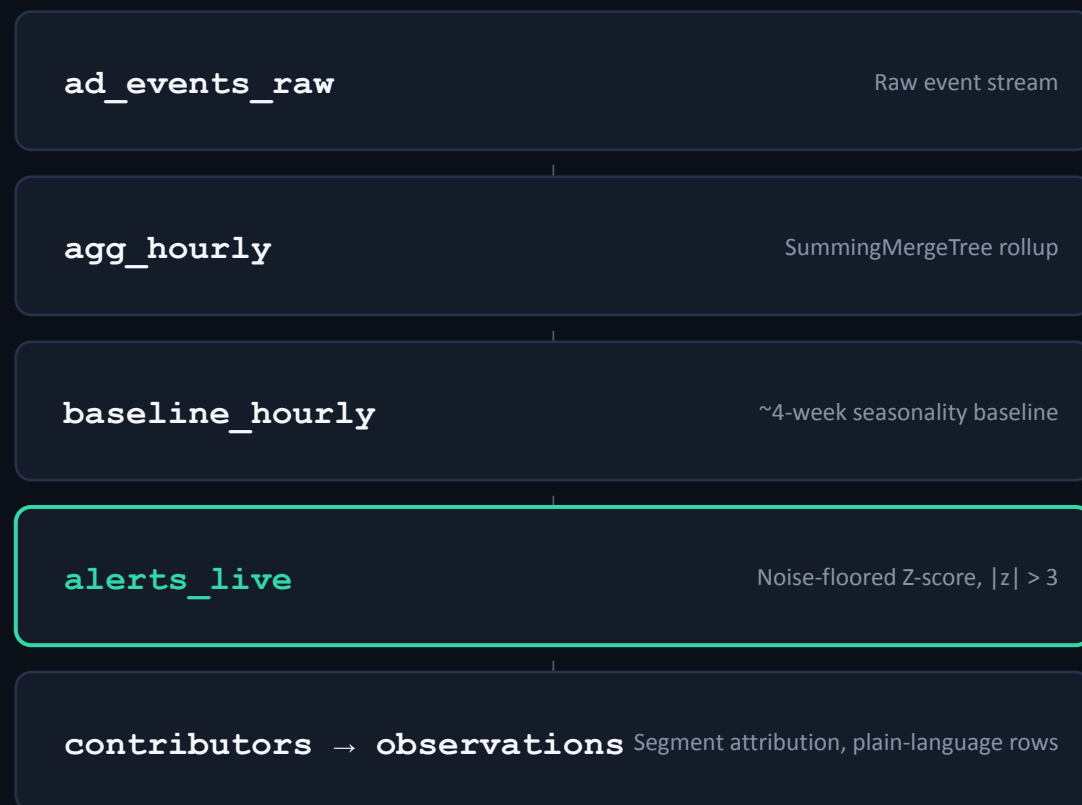
5

Diagnosis + chat

Deterministic citations become plain English; follow-up chat stays grounded in the same evidence.

ARCHITECTURE

Analysis lives in ClickHouse. The LLM never drills the data.



Hosted: Vercel (web) · Railway (api + RCA) · ClickHouse Cloud · Gemini API · Langfuse JP Cloud

APPLICATION LAYER

- **Node API (RCA engine)**
Reads the view layer, packages diagnosis + trace
- **Gemini (flash-lite)**
Narrates evidence JSON only — never sees raw events
- **Langfuse**
OTEL spans on chat / investigate / narrate
- **React / Vite**
Alerts wall, investigation workspace, chat

One click from alert to root cause

Metric tree

Revenue decomposed into Requests × Fill rate × Render rate × eCPM — culprit vs ruled-out at a glance.

Cited diagnosis

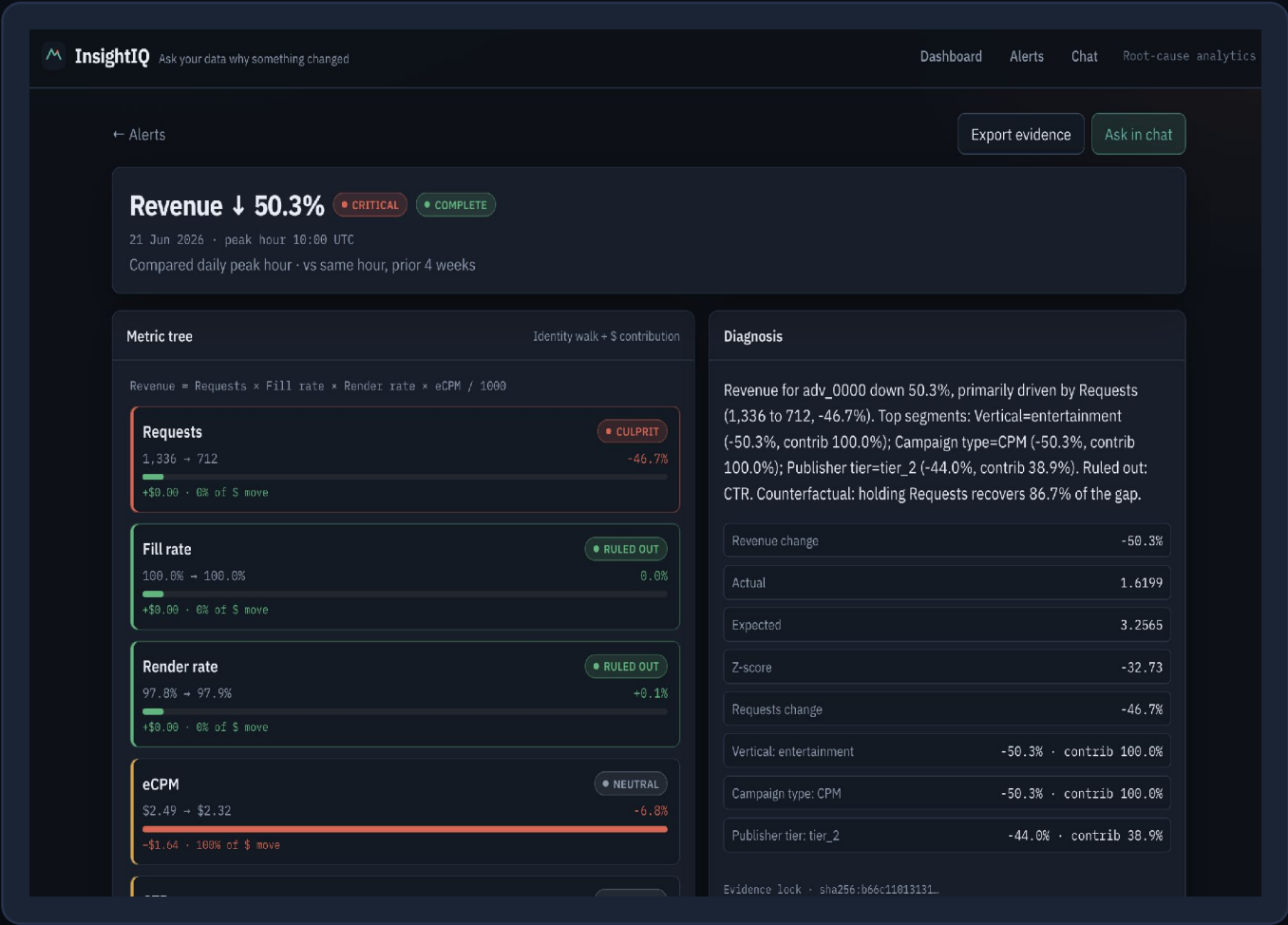
Every number — actual, expected, Z-score, segment contribution — traces to a ClickHouse row.

Counterfactual

“Holding Requests recovers 86.7% of the gap” — the answer, not just the alert.

Export evidence

SHA-256 evidence lock + ordered trace[] for a fully auditable investigation.



WHY IT WINS

Built for trust at InMobi scale

1

Analysis in the warehouse

ClickHouse does every computation. Low egress, no raw-event scans reaching the LLM, high trust by design.

2

Explainable, not black-box

Deterministic seasonality baselines and contribution analysis beat opaque ML for an ops audience.

3

Ready for the unseen incident

Export bundles, SHA-256 evidence hash, and ordered trace[] — graded on incidents the team never saw.

~4 weeks

seasonality baseline window

$|z| > 3$

alert threshold

SHA-256

evidence lock per investigation

THANK YOU

404Duos

InsightIQ — from alert to answer



Vishnu Bhagwat

@rogue-wild



Sethukumar J

@SethukumarJ

TRY IT

Live app

`insight-iq-woad.vercel.app`

API health

`insightiq-production-be0e.up.railway.app/h
ealth`

Repo

`github.com/rogue-wild ·
github.com/SethukumarJ`

Demo video

`2-3 min walkthrough — linked in README`